Project: Mini-Project

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Course:	GPU Computing

Contents

1 Intro

Throughout all parts of the mini-project, it was chosen to compare two **operations** namely: addition & multiplication. To start, a comparison between the operations will be done, as requested in part 1. Following that, the kernel will be updated to execute on multiple elements simultaneous (one-to-many) mapping.

For the third part, the compute intensity will be increased by applying a loop count factor over the computations.

The final part will compare the local vs global memory location, as indicated in the scheme[1] in the assignment.

2 Structure

TODO

3 Methodology

TODO

[1] http://parallel.vub.ac.be/education/gpu/labs/miniproject_step4_with_local%20memory.jpg

4 Part 1: Addition vs Multiplication

4.1 Setup

TODO

4.2 Memory Bandwidth

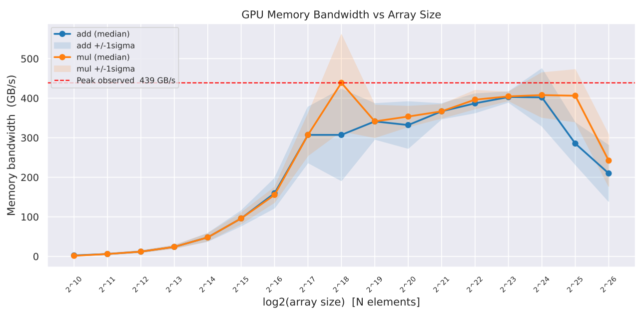

Figure 1:

TODO

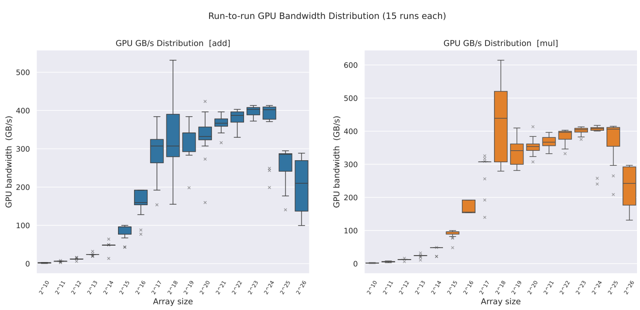

Figure 2:

4.3 Compute Throughput

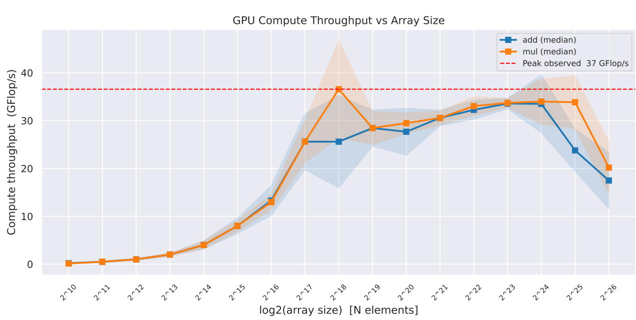

Figure 3:

5 Part 2: Elements Per Thread

6 Part 3: Roofline Model

7 Part 4: Local vs Global

8 Extra: Desktop vs Macbook

9 Appendix

9.1 Part 1

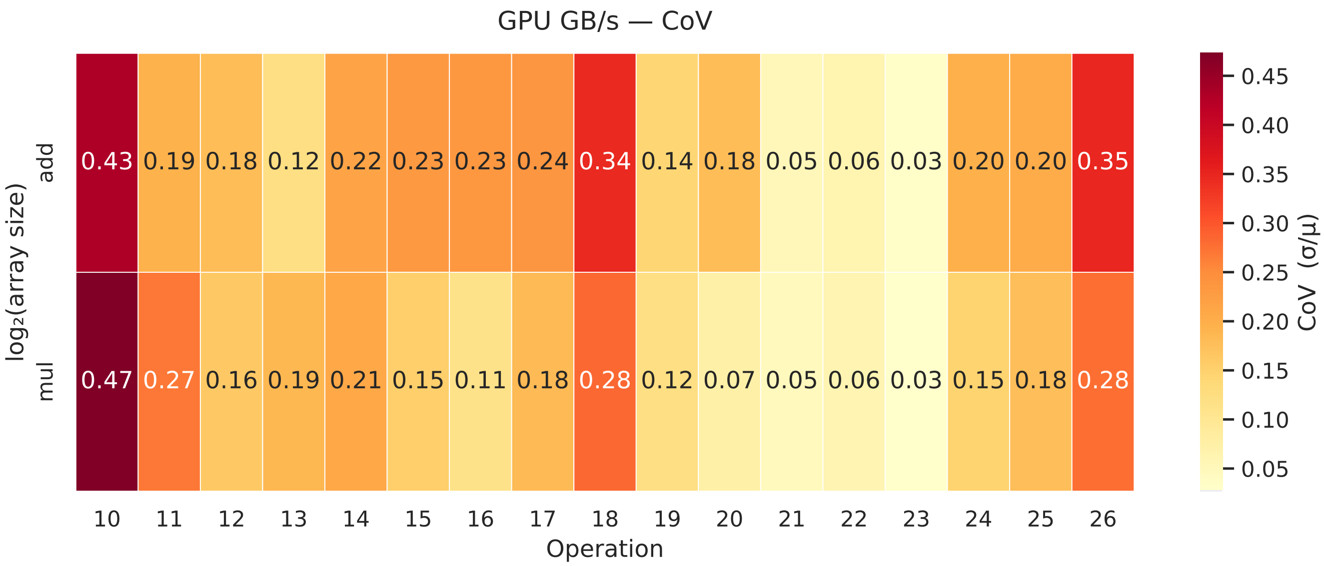

Figure 4:

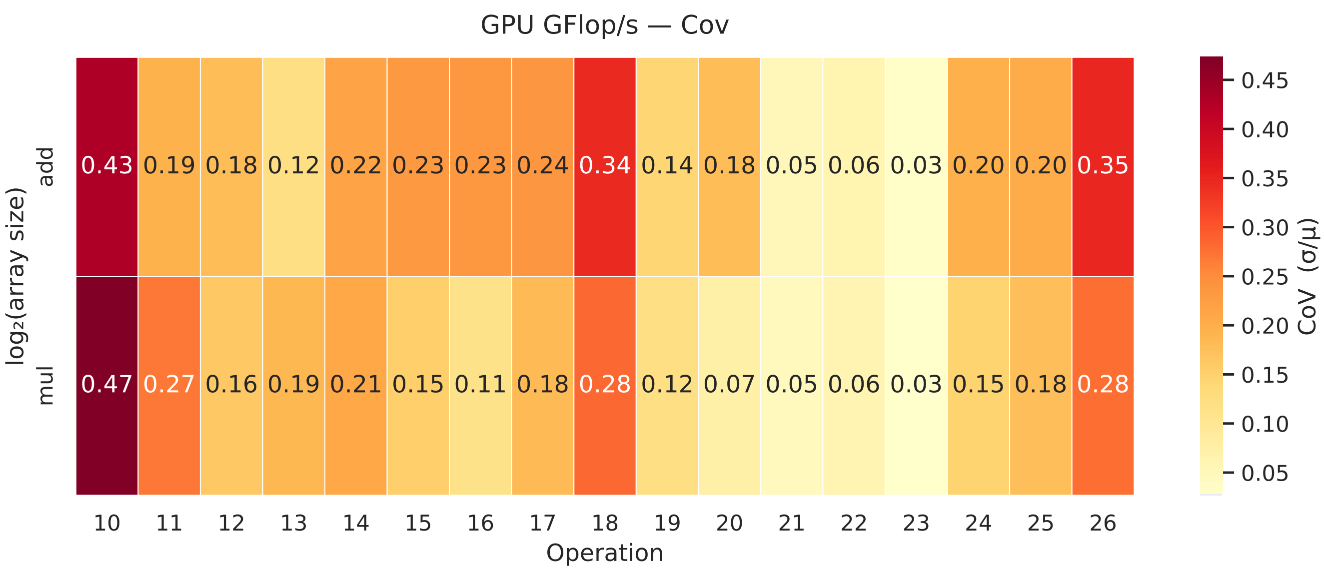

Figure 5:

9.2 Specifications

9.2.1 Macbook

Part	Value
CPU	M2 Pro (6 performance and 4 efficiency)
RAM	16GB
OS	**TODO**

Table 1: Macbook Specifications

9.2.2 Desktop

Part	Value
CPU	Ryzen 9 5950X
GPU	RTX 3070
OpenCL	**TODO**
Driver Version	**TODO**
RAM	64GB (3200 Mhz)
OS	Windows Version 10.0.22631 Build 22631

Table 2: Desktop Specifications

Bibliography